Experiment - 06

Workflow Automation using Deluge for a Business Application in Zoho Creator

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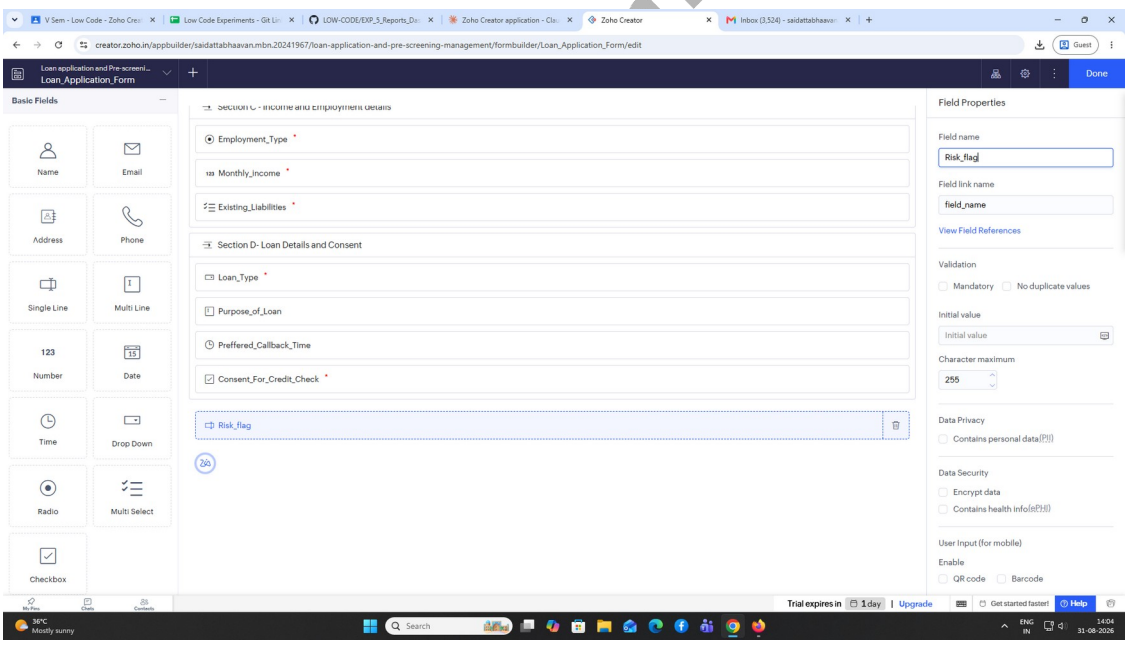
AIM

To automate a business rule on Loan_Application_Form using Zoho Creator's Deluge scripting language, so that every submitted loan application is automatically evaluated and tagged with a risk classification — without requiring a loan officer to manually review each submission before an initial triage decision is made.

PROCEDURE

1. Adding the Risk_flag Field

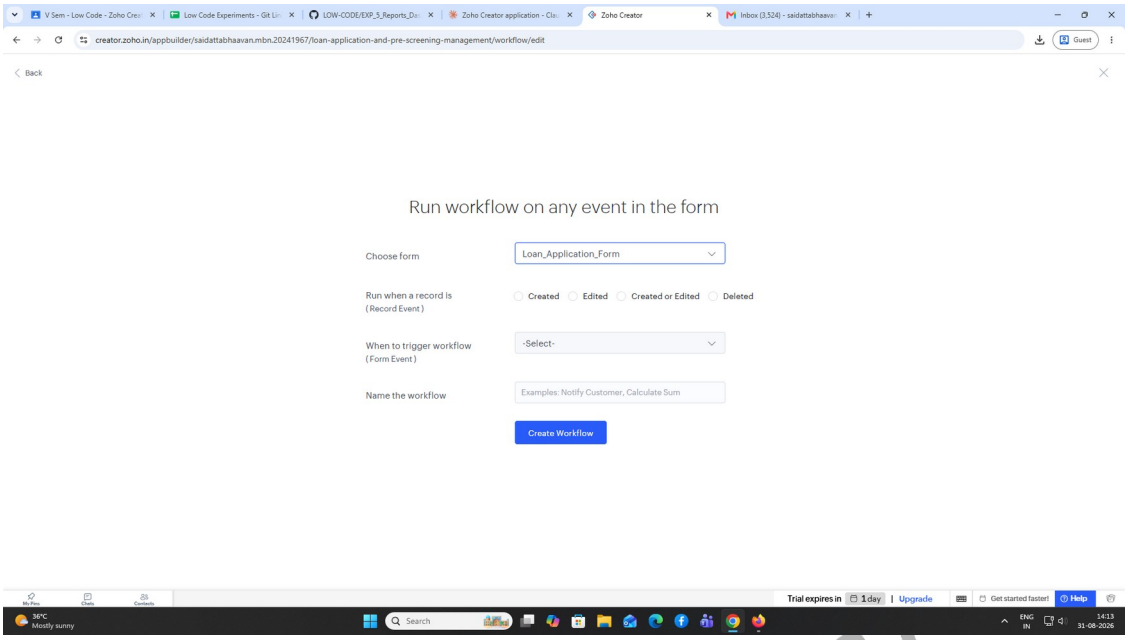
A new Single Line field, Risk_flag, was added to Loan_Application_Form via the Form Builder, positioned after the existing Section D fields. This field was left non-mandatory, since its value is intended to be set automatically by a workflow script rather than typed by the applicant.



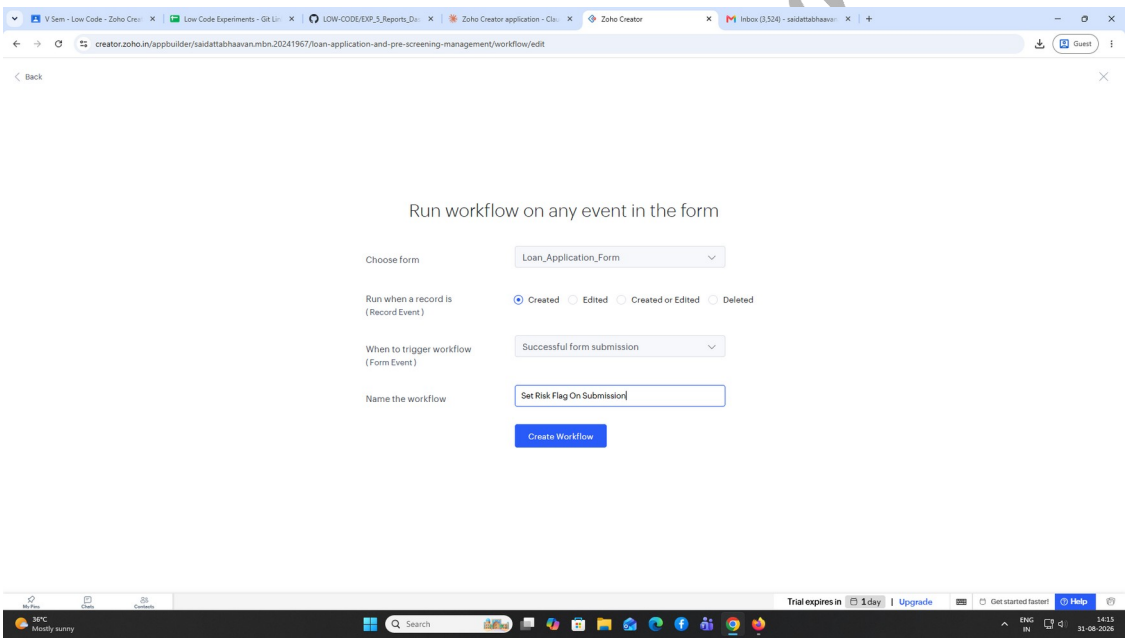
Risk_flag field added to Loan_Application_Form via the Form Builder

2. Creating the Workflow Rule

A new workflow rule was created from the app's Workflow tab. The rule was configured to run on Loan_Application_Form, triggered when a record is Created, specifically on Successful form submission, and named “Set Risk Flag On Submission”.



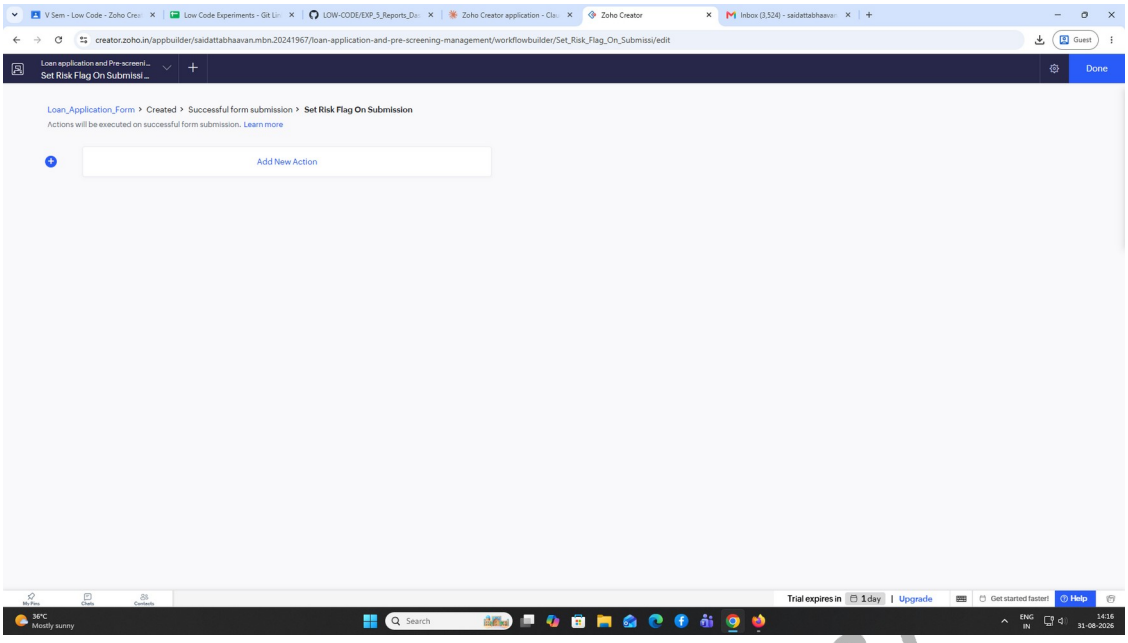
Workflow creation screen — form and trigger configuration



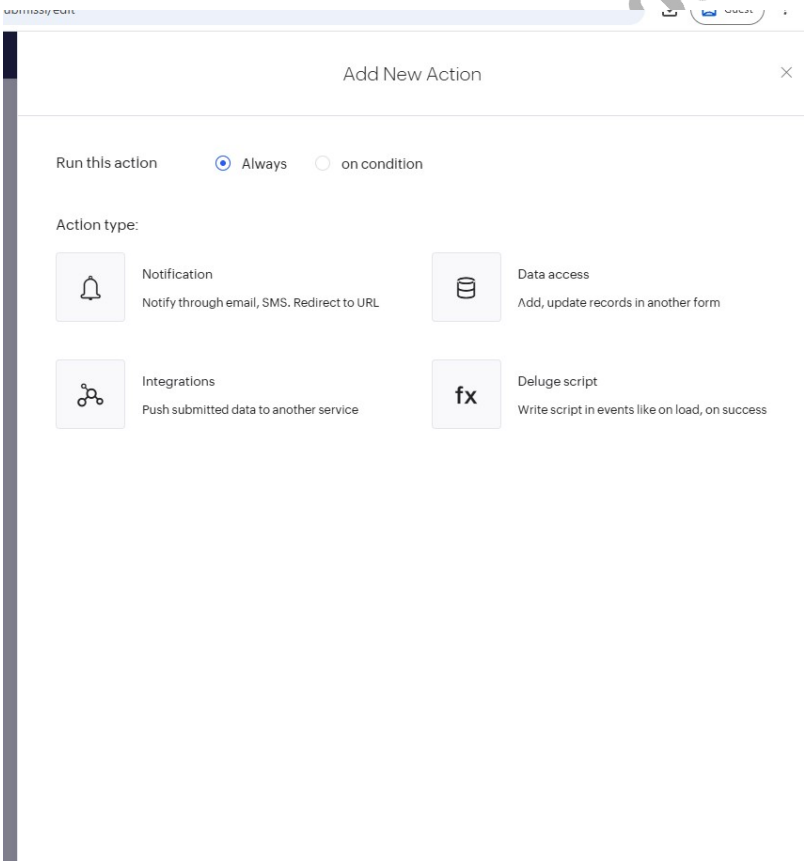
Workflow configured: Loan_Application_Form, Created, Successful form submission

2.1 Adding the Action

On the newly created workflow's action list, a new action was added and configured as a Deluge script, since the required logic (a conditional if/else check) needed genuine branching rather than a simple field-update rule.



Empty action list for the newly created workflow



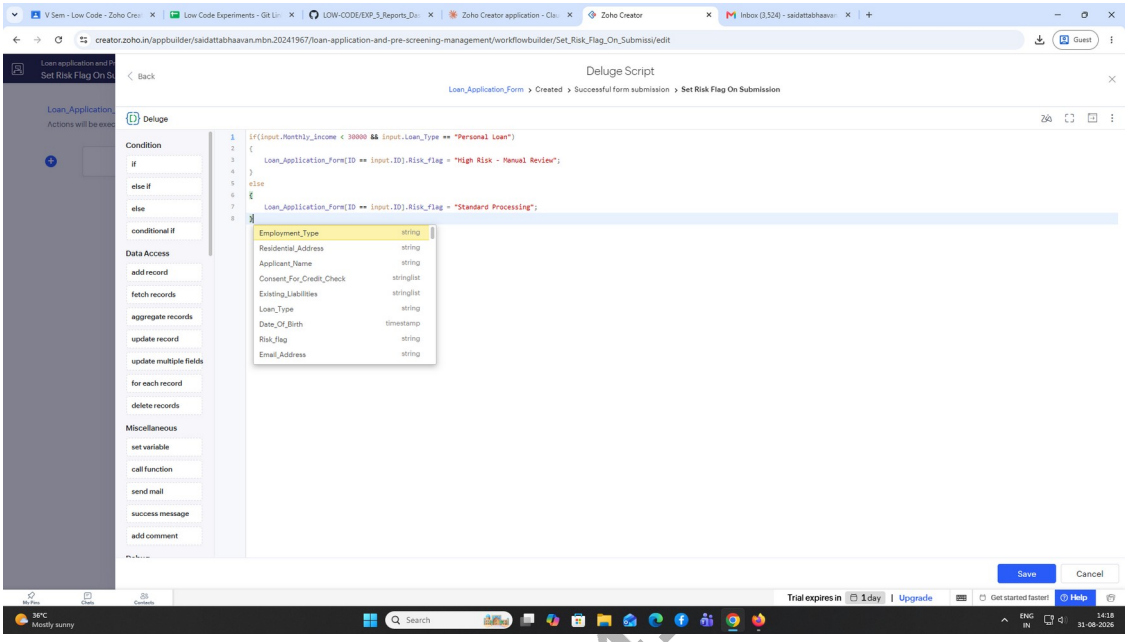
Add New Action dialog — Deluge script action type selected

3. Writing the Deluge Script

The intended logic was: if an applicant's Monthly_income is below 30,000 and their Loan_Type is “Personal Loan”, the application is flagged High Risk and routed for manual review; otherwise, it is marked as Standard Processing.

3.1 Initial Script and Debugging

An initial version of the script referenced the submitted record indirectly, using an input variable and a lookup expression of the form Loan_Application_Form[ID == input.ID].Risk_flag. This produced an "Improper Statement Error" on save, since the workflow script executes directly in the context of the record that triggered it — an explicit input prefix and ID-based lookup are unnecessary and syntactically invalid in this context.



Initial Deluge script using an input./ID lookup pattern that produced a syntax error on save

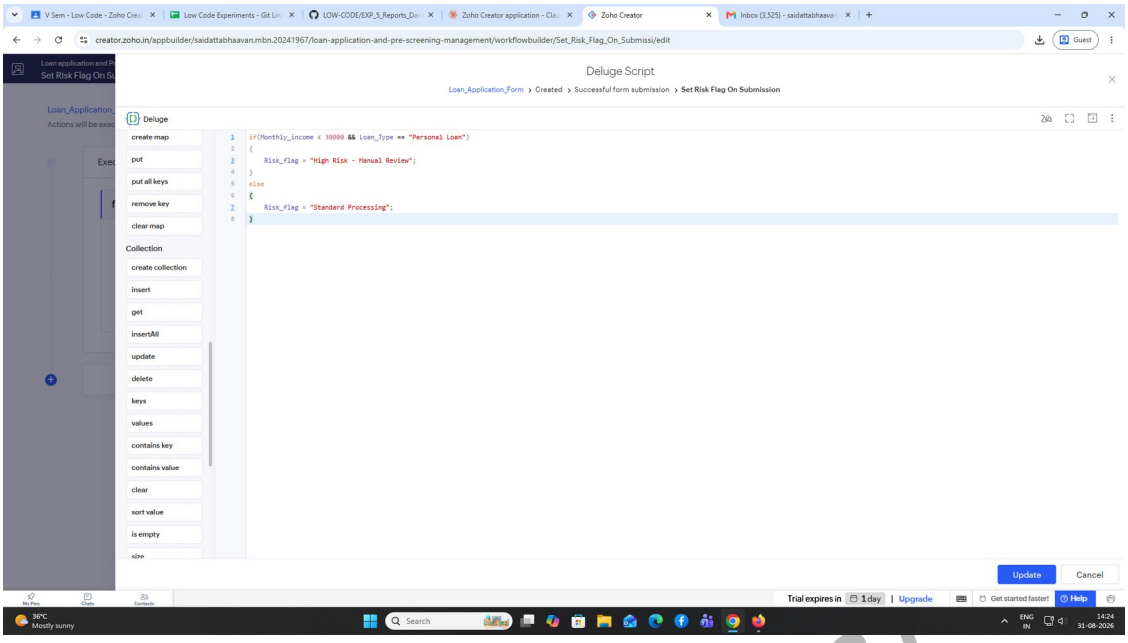
3.2 Corrected Script

The script was rewritten to reference the current record's fields directly, without the input prefix or lookup expression:

```
if(Monthly_income < 30000 && Loan_Type == "Personal Loan")
{
	Risk_flag = "High Risk - Manual Review";
}
else
{
	Risk_flag = "Standard Processing";
}
```

This version saved without error, since Deluge allows direct field access on the record that is currently being processed by the workflow.

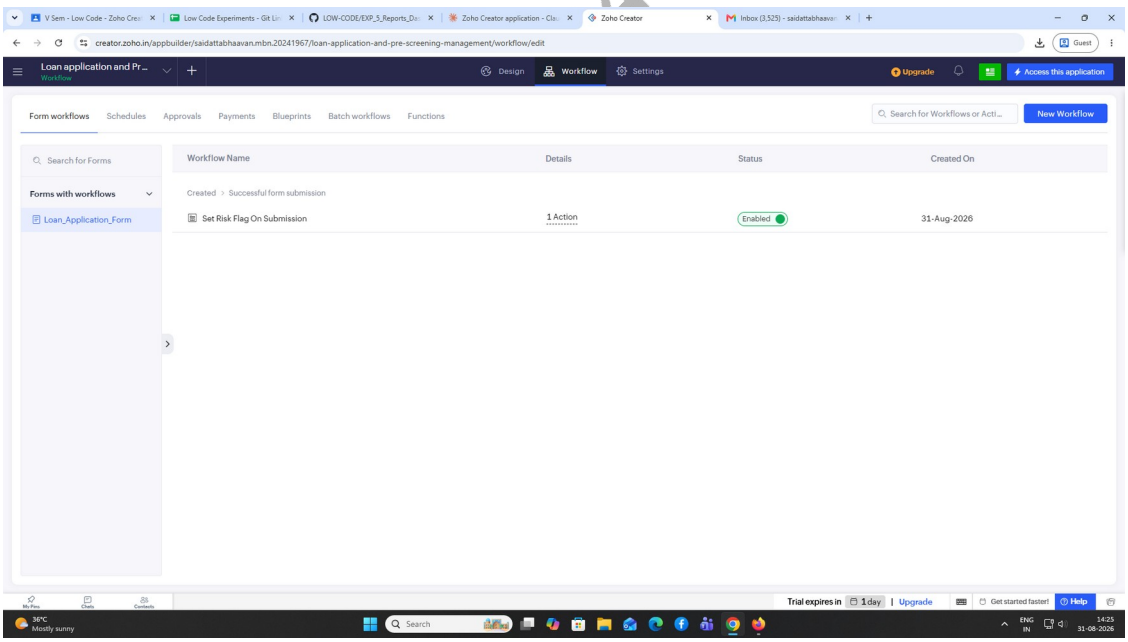
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Corrected Deluge script, saved without error

4. Enabling and Verifying the Workflow

After saving the corrected script, the workflow was confirmed on the Workflow tab as "Set Risk Flag On Submission", listed with 1 Action and a status of Enabled, tied to the Created → Successful form submission event on Loan_Application_Form.

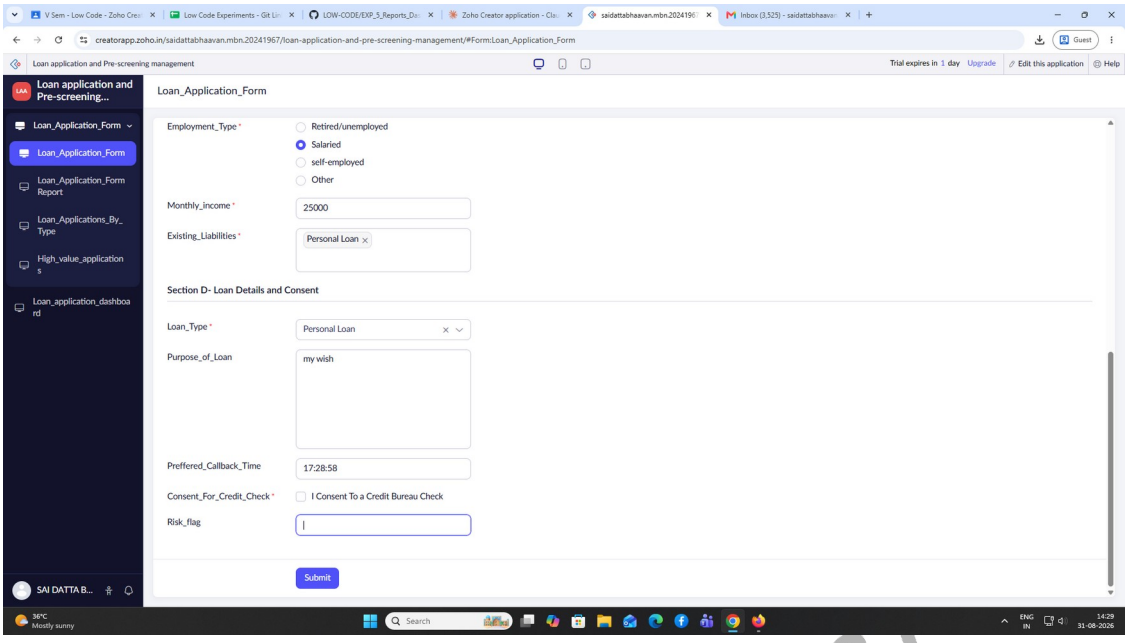


Set Risk Flag On Submission workflow shown as Enabled with 1 Action

5. Testing the Workflow

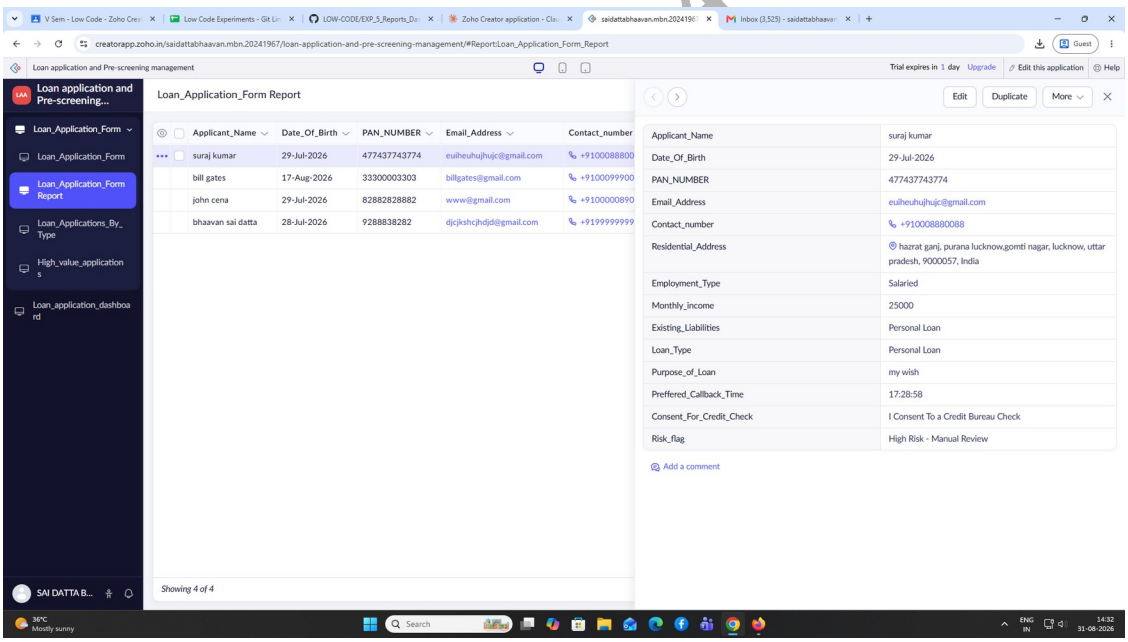
5.1 Test Case 1 — High Risk Branch

A test application was submitted via the live form with Monthly_income set to 25,000 and Loan_Type set to Personal Loan — satisfying both conditions of the if branch.



Live form submission for Test Case 1 — Monthly_income 25000, Loan_Type Personal Loan

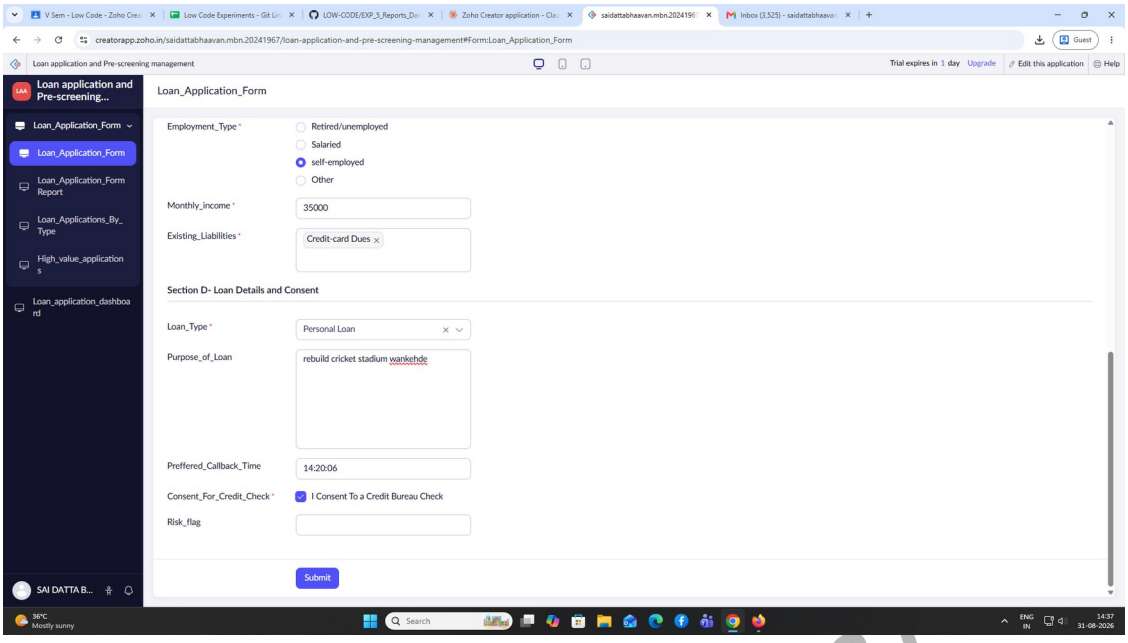
Upon checking the submitted record (applicant suraj kumar) in Loan_Application_Report, the Risk_flag field correctly showed “High Risk - Manual Review”, confirming the if branch executed as intended.



Loan_Application_Report detail view — Risk_flag correctly set to “High Risk - Manual Review”

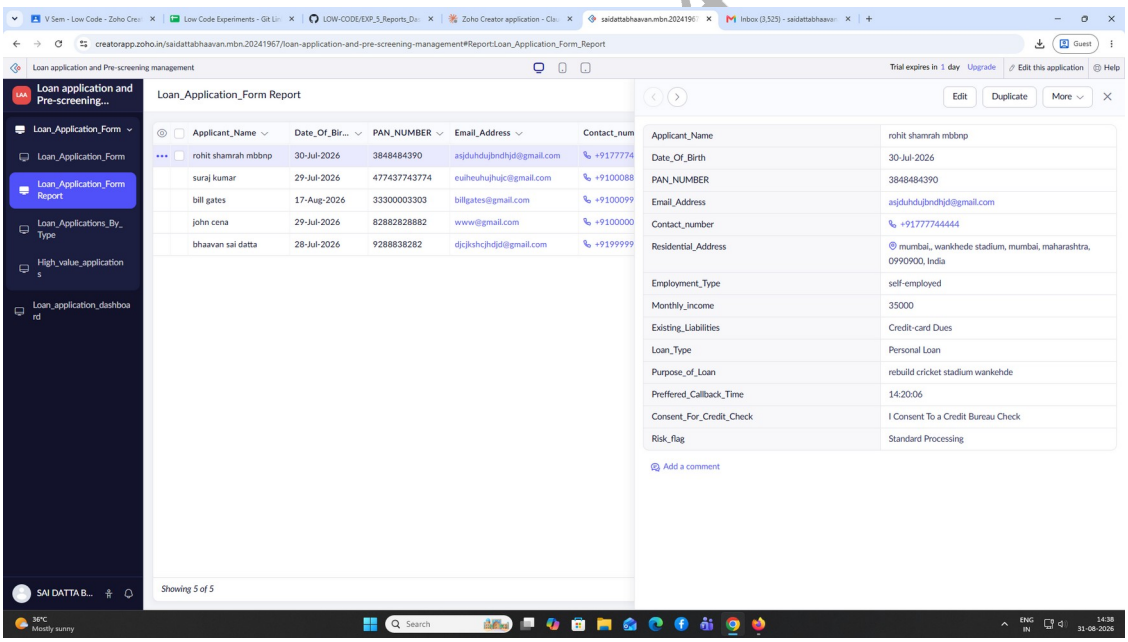
5.2 Test Case 2 — Standard Processing Branch

A second test application was submitted with Monthly_income set to 35,000 (above the 30,000 threshold) and Loan_Type set to Personal Loan — satisfying only one of the two required conditions, which should route it to the else branch.



Live form submission for Test Case 2 — Monthly_income 35000, Loan_Type Personal Loan

Upon checking the submitted record (applicant rohit shamrah mbbnp) in Loan_Application_Report, the Risk_flag field correctly showed “Standard Processing”, confirming the else branch executed correctly when the income condition was not met.



Loan_Application_Report detail view — Risk_flag correctly set to “Standard Processing”

RESULT

A Deluge workflow script was successfully written and deployed on Loan_Application_Form to automatically classify each submitted loan application as either “High Risk - Manual Review” or “Standard Processing”, based on the applicant's Monthly_income and Loan_Type. An initial version of the script using an input-based record lookup produced a syntax error, which was resolved by referencing the current record's fields directly, appropriate for a workflow executing in the context of the triggering record. Both branches of the conditional logic were independently tested and verified against real submitted records, confirming the automation behaves correctly. The exercise thus demonstrated an

understanding of Zoho Creator's Deluge scripting language, workflow trigger configuration, and the debugging of common script errors.

ROLL NO: 241401087